12 A particle is oscillating with simple harmonic motion described by the equation:

$$y = 5\sin(20\pi t)$$

where y is in meters and t is in seconds.

How long does it take the particle to travel from its position of maximum displacement to a position half way between the maximum displacement and the equilibrium position?

A Less than 12.5 ms

B 12.5 ms

C Between 12.5 ms and 25 ms

D 25 ms